

A Coherence Triangle Linking the Initial State, the Origin of the Microwave Background, and the Missing Mass Problem

CMB as Latent Heat of Vacuum Birth

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Abstract

Three outstanding problems in cosmology — the nature of the initial state, the origin of the cosmic microwave background, and the missing mass problem — are conventionally treated as independent. We show that within the LGQV framework (Local Gravity of Quantum Vacuum), they form a closed logical triangle in which each vertex supports the other two.

The K-Limit — the maximum density at which the chiral condensate exists — eliminates the singularity. The initial state is a three-sphere S^3 of radius ≈ 15 AU filled with quark-gluon-vacuum plasma (QGVP): a unified substance in which the separation into "matter" and "vacuum" has not yet occurred. At the QCD phase transition ($T_c \approx 155$ MeV), QGVP separates into three components: baryonic matter, vacuum energy ($w = -1$), and thermal radiation — the CMB. The CMB is thus the latent heat of vacuum birth, with baryon-to-photon ratio $\eta = \alpha^4 = 6.20 \times 10^{-10}$.

This changed photon production mechanism recalibrates the baryon fraction upward from the standard $\Omega_b = 0.049$. Combined with a top-heavy initial mass function at low metallicity — producing trillions of invisible compact remnants over 13 Gyr — the total baryonic mass approaches the "missing mass" inferred from rotation curves. The spatial distribution (flat rotation curves, core profiles, baryonic Tully-Fisher relation) is produced by the σ -term shift of the chiral condensate, which follows the gravitational potential rather than concentrating at the centre: mass from dead stars, profile from vacuum.

No vertex can be independently proven. Each is confirmed by the consistency of the other two. The full bookkeeping from today back to the K-Limit is self-consistent: the Friedmann integral gives 13.48 Gyr, matching observation within 2.3%. An additional cooling factor of ~ 440 relative to standard adiabatic expansion reflects entropy injection by the vacuum during the QGVP $\rightarrow$ matter + vacuum transition.

Keywords: cosmological constant, chiral condensate, cosmic microwave background, missing mass, initial mass function, quark-gluon-vacuum plasma, K-Limit, coherence

1. Introduction: Three Islands

Standard cosmology treats three problems separately.

The initial state. The universe begins with a singularity — infinite density, zero volume, undefined physics. General relativity breaks down. Quantum gravity is invoked but not available. The singularity is not a prediction; it is a confession of ignorance.

The cosmic microwave background. The CMB is interpreted as thermal radiation released at recombination ($z \approx 1100$). The baryon-to-photon ratio $\eta \approx 6.1 \times 10^{-10}$ determines the baryon fraction: $\Omega_b = 0.049$, approximately 5% of the total energy budget.

The missing mass. The remaining matter budget is assigned to cold dark matter particles: $\Omega_{dm} \approx 0.266$. This component has never been directly detected despite four decades of experimental search.

These problems are linked only by the Λ CDM parameter fit. One can change the theory of dark matter without touching the CMB interpretation. One can modify the initial state without affecting the dark matter budget. They are three islands connected by a bridge of parameters, not by physics.

We show that in the LGQV framework, these three islands are one continent.

2. The Framework: One Action

The LGQV framework is generated by a single action:

$$S = \int d^4x \sqrt{(-g)} \left\{ \frac{(m^2_{PI})}{2} [R + R^2 / (6M^2)] + (1 + 2\alpha) \mathcal{L}_m \right\}$$

The $R^2/(6M^2)$ term is the Starobinsky term [1], where $M \approx 3 \times 10^{13}$ GeV is the scalaron mass fixed by CMB amplitude [2]. It endows spacetime with elastic stiffness.

The factor $(1 + 2\alpha)$ is the vacuum–matter coupling:

$$\alpha = f_b \times (\sigma_{\pi N} + \sigma_s) / (3 \times m_N) = 0.005$$

derived from measured QCD sigma terms [3,4] with zero free parameters. The equilibrium vacuum does not gravitate; only the shift of the chiral condensate in the presence of baryonic matter produces local curvature. This shift is a Lorentz scalar with $w = -1$ [5].

The cosmological constant follows:

$$\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m^4_{Pl} = 1.8 \times 10^{-52} \text{ m}^{-2}$$

Observed: $1.1 \times 10^{-52} \text{ m}^{-2}$. Ratio: 1.65. Zel'dovich missed by 10^{120} . This formula misses by 1.65.

The dynamical component — how vacuum energy density depends on the Hubble rate $H(t)$ — has been independently derived by Zhitnitsky [6,7] from QCD topological sectors with nontrivial holonomy: $\rho_{DE}(t) \propto H(t)$. Barvinsky and Zhitnitsky [8] demonstrated that this contribution is physical and cannot be removed by UV renormalization, because it has topological infrared origin. Our Λ_0 provides the asymptotic value toward which H evolves; the Zhitnitsky mechanism provides the dynamical approach to that value.

3. Vertex A: The Initial State

3.1 The K-Limit

The K-Limit is the maximum density at which the chiral condensate can exist: $\rho_K \approx 2.3 \times 10^{17} \text{ kg/m}^3$.

As density increases, baryons approach each other and their σ -term clouds overlap. The condensate shift grows. The energy of the shifted condensate — a Lorentz scalar with $w = -1$ — grows as ρ^2 . Gravitational compression grows as ρ . At nuclear density, the repulsive pressure of the shifted condensate equals the gravitational compression. Further collapse is impossible.

The chiral phase transition is not speculative. It is studied on the lattice, observed in heavy-ion collisions at RHIC and the LHC, and occurs at $T_c \approx 155 \text{ MeV}$.

3.2 Quark-Gluon-Vacuum Plasma (QGVV)

At the K-Limit ($T > T_c$), the chiral condensate is restored: $\langle \bar{q}q \rangle \approx 0$. There are no protons, no neutrons, no "vacuum" as a separate entity. The contents of S^3 constitute a single unified substance — quark-gluon-vacuum plasma (QGVV) — in which the separation into "matter" and "vacuum" has not yet occurred.

This is not standard QGP as observed at RHIC and LHC. Standard QGP is a small fireball (10^{-14} m , 10^{-23} s) embedded in ordinary vacuum with an intact chiral condensate outside. QGVV fills the entire S^3 with no boundary to ordinary vacuum. There is no "outside." 57% of its mass-

energy is a component that will become vacuum energy upon phase separation, but which currently exists as an undifferentiated part of the plasma.

A neutron star has similar density ($\sim 2 \times 10^{17} \text{ kg/m}^3$) but is cold ($T \ll T_c$). The condensate exists. Nucleons exist. The matter-vacuum separation occurred billions of years ago. The neutron star contains negligible vacuum energy density relative to its baryonic density. The QGVP is the opposite: hot, deconfined, unseparated, with 57% vacuum component.

The properties of QGVP on a closed S^3 of finite volume are computable in lattice QCD but have not been computed. Standard lattice calculations use a flat torus T^3 with periodic boundary conditions. The S^3 topology introduces nontrivial features: $\pi_3(\text{SU}(3)) = \mathbb{Z}$ provides topological sectors that may constrain the θ -vacuum geometrically.

3.3 The initial state

The K-Limit replaces the singularity. The initial state is a closed three-sphere S^3 filled with QGVP at nuclear density. Its radius is determined by the full bookkeeping (Section 6): **$R_{\text{initial}} \approx 15 \text{ AU}$** . Finite volume. Finite density. Finite temperature.

3.4 Why no gravitational collapse

At nuclear density the total mass-energy is $5.25 \times 10^{55} \text{ kg}$. The Schwarzschild radius for this mass is $R_s \approx 8 \text{ Gly}$ — ten quadrillion times larger than R_{initial} . Standard logic suggests instant collapse. Three mechanisms prevent it.

Closed topology. The Schwarzschild solution assumes asymptotically flat space with a distinguished centre. On S^3 , no centre exists. Every point is equivalent to every other. Gravity pulls equally in all directions. The net force on any element is zero — not because gravity is absent, but because it is symmetric. Applying the Schwarzschild radius to a homogeneous closed universe is a category error.

Dominant $w = -1$ component. At the K-Limit, 57% of mass-energy has $w = -1$ (the vacuum component of QGVP). The Friedmann acceleration equation gives $\ddot{a}/a = -(4\pi G/3)(\rho + 3p)$. For $w = -1$: $\rho + 3p = -2\rho < 0$. The acceleration is positive — expansion, not collapse. The universe is already accelerating at birth.

Homogeneity. Collapse requires density fluctuations. At the K-Limit, the QGVP is thermalised. The speed of sound $c_s = c/\sqrt{3} \approx 0.58c$ — relativistic pressure resists any compression. Macroscopic inhomogeneities are smoothed on timescales shorter than the Hubble time.

3.5 Origin of inhomogeneities

The homogeneity is not absolute. Three sources of fluctuations seed the future cosmic web.

Quantum fluctuations of the scalaron. The R^2 -term is a dynamical degree of freedom — a scalar field with mass $M = 3 \times 10^{13} \text{ GeV}$. As the membrane relaxes, its zero-point fluctuations

are stretched to macroscopic scales, producing a nearly scale-invariant spectrum. The predicted spectral index $n_s = 0.964$ matches the observed $n_s = 0.965 \pm 0.004$ [2].

QCD phase transition fluctuations. At $T_c \approx 155$ MeV, QGVP separates into matter + vacuum. Phase transitions are inherently inhomogeneous — the condensate nucleates at random points, bubbles grow and merge.

Thermal fluctuations. At $T \sim 10^{12}$ K, thermal fluctuations $\delta\rho/\rho \sim 1/\sqrt{N}$ are present but subdominant ($\delta\rho/\rho \sim 10^{-29}$ on Hubble scales).

The dominant source is the first: quantum fluctuations of the scalaron, stretched by the membrane relaxation.

3.6 The double lock

The minimum radius of S^3 is constrained by two independent mechanisms: the K-Limit (nuclear physics prevents compression beyond ρ_K) and the membrane stiffness (the R^2 -term resists deformation). Both yield ~ 15 AU. Two locks on one door, both engaged at the same threshold.

4. Vertex B: The CMB as Latent Heat of Vacuum Birth

4.1 The phase transition

At $T_c \approx 155$ MeV, the QGVP undergoes a phase transition. Three things happen simultaneously:

Quarks are confined into hadrons — protons and neutrons appear.

The chiral condensate $\langle \bar{q}q \rangle$ forms — the "medium" comes into existence.

The vacuum component separates — 57% of mass-energy transitions into a distinct phase with $w = -1$.

This separation releases latent heat. When water freezes, it releases 334 J/g. When QGVP separates into matter + vacuum, it releases the energy of the transition. This energy thermalises into photons. These photons are the CMB.

4.2 $\eta = \alpha^4$

The baryon-to-photon ratio is determined by the four-vertex process: quark $\rightarrow$ condensate $\rightarrow$ condensate $\rightarrow$ photon. Each vertex contributes a factor α .

$$\eta = \alpha^4 = (0.005)^4 = 6.20 \times 10^{-10}$$

Observed: 6.12×10^{-10} . Agreement: 1.2%.

This is not baryogenesis. No CP violation is needed. No Sakharov conditions are invoked. On a closed manifold S^3 there is no "moment before" the universe — no symmetric initial state from which an asymmetry must be generated.

4.3 The factor of 440

Standard cosmology predicts $T \propto 1/a$ (adiabatic expansion, conserved photon number). This gives $R_{\text{initial}} = R_{\text{now}} \times T_{\text{now}}/T_{\text{initial}} = 6600 \text{ AU}$. But the bookkeeping gives $R_{\text{initial}} = 15 \text{ AU}$. The ratio: 440.

This factor is the signature of vacuum radiation. In the LGQV framework, photons are produced by the vacuum–matter interaction during the phase separation. Entropy is injected. The universe cools faster than $1/a$.

$440^3 \approx 8.5 \times 10^7$ — the vacuum produced ~ 85 million times more photons than were present in the initial QGVP. This is testable: spectral distortions (μ -distortion, y -distortion) should differ from the standard prediction. PIXIE can measure this.

4.4 Nucleosynthesis

At the K-Limit, $T > T_c \approx 155 \text{ MeV}$. At this temperature, protons and neutrons do not exist. Matter is QGVP. Thermonuclear reactions require nucleons as bound states. No nucleons — no reactions. This is why the initial state does not produce heavy elements despite its nuclear density.

After the phase transition ($T < T_c$), nucleons form. But thermal photons photodissociate any nucleus heavier than a proton until $T \sim 0.1 \text{ MeV}$ ($t \sim 3 \text{ min}$), when the deuterium bottleneck opens and standard BBN proceeds. The result: 25% ^4He , 75% H, traces of D and ^7Li — identical to standard BBN, because $\eta = \alpha^4 = 6.20 \times 10^{-10}$ matches the standard value, and nuclear reactions do not depend on cosmological models.

No heavy elements are produced because: (i) no stable nuclei exist at $A = 5$ and $A = 8$ (the mass gaps), and (ii) the universe expands too fast for the triple-alpha process, which requires high helium density sustained over long timescales — conditions found in stellar cores but not in the expanding universe.

The 1% modification $G_{\text{eff}} = G(1 + 2\alpha)$ shifts ^4He yield by $\sim 0.1\%$ — undetectable with current precision.

5. Vertex C: The Missing Mass — Dead Stars and Vacuum Profile

5.1 The standard calibration is model-dependent

The baryon fraction $\Omega_b = 0.049$ is extracted from CMB acoustic peaks assuming: (i) photons are annihilation remnants, (ii) gravity is standard GR, (iii) dark matter exists as particles. In the LGQV framework, all three assumptions are violated. The calibration must be redone.

5.2 Top-heavy initial mass function

The standard IMF (Kroupa/Salpeter) is calibrated on present-day star formation at solar metallicity. Extrapolating it to the first 3–5 Gyr is unjustified for three reasons.

Low metallicity. Without metals (C, O, Fe), gas cannot cool efficiently and fragments into fewer, more massive clumps. At $Z < 0.001 Z_\odot$, typical stellar mass shifts from $\sim 0.3 M_\odot$ to $10\text{--}100 M_\odot$ [9,10]. JWST confirms excess luminosity at $z > 10$, consistent with top-heavy IMF [11].

Reduced fragmentation. Higher Jeans mass at low metallicity means fewer fragments, each more massive. Instead of 1000 red dwarfs — 10 massive stars, each living 3–10 Myr and leaving a compact remnant.

Higher vacuum density in LGQV. At earlier epochs the S^3 was smaller, vacuum energy density higher (same energy, smaller volume — membrane geometry). This increases the effective Jeans mass further — a factor unique to LGQV.

5.3 The cemetery of dead stars

Over 13 Gyr with top-heavy early IMF, a typical galaxy accumulates:

Component	Mass ($M_\odot$)
Living stars (visible)	5×10^{10}
Stellar black holes	$10^{11} - 1.5 \times 10^{12}$
Neutron stars	7.5×10^{10}
White dwarfs	6×10^{10}
Gas (HI + H ₂ + CGM)	5×10^{10}
Total	$2 \times 10^{11} - 1.8 \times 10^{12}$

The standard "missing mass" of a Milky Way analogue from rotation curves is $1\text{--}2 \times 10^{12} M_\odot$. In the aggressive scenario, the invisible baryonic mass alone approaches this value.

5.4 Mass from dead stars, profile from vacuum

Dead stars provide the mass but not the distribution. Compact remnants concentrate toward the disc. Rotation curves require mass distributed in an extended halo.

The vacuum provides the profile. Every baryon — living or dead — shifts the chiral condensate. Through vacuum capture, this shift is retained by the gravitational potential of the galaxy, forming an extended halo of shifted condensate with $w = -1$.

The density profile of the shifted condensate follows the gravitational potential ($\sim 1/r$ at large r), giving $v(r) \approx \text{const}$ — flat rotation curves. In the centre, baryons dominate and the vacuum contribution is subdominant — a core, not a cusp, solving the cusp-core problem.

The baryonic Tully-Fisher relation ($M_{\text{total}} \propto v^4$) is natural: more baryons $\rightarrow$ more condensate shift $\rightarrow$ more vacuum halo. The scaling is determined by α .

Void galaxies should retain the same $M_{\text{total}}/M_{\text{visible}}$ ratio as cluster galaxies — because the effect is local (determined by the galaxy's own baryons), not environmental. If the missing mass were particles, void galaxies would be dark-matter-poor.

6. Full Bookkeeping

6.1 Today's inventory

Volume of S^3 from $\Omega_K = -0.044$: $R_{\text{now}} = 69 \text{ Gly}$, $V_{\text{now}} = 5.5 \times 10^{81} \text{ m}^3$.

Component	Mass-energy (kg)	Fraction
Matter ($\Omega_m = 0.315$)	1.65×10^{55}	31.5%
Vacuum ($\Omega_\Lambda = 0.685$)	3.60×10^{55}	68.5%
Radiation	4.8×10^{51}	$\sim 0\%$
Total	5.25×10^{55}	

6.2 Packing at the K-Limit

All contents compressed to $\rho_K = 2.3 \times 10^{17} \text{ kg/m}^3$:

Component	Volume at ρ_K	Fraction
Matter	$7.0 \times 10^{37} \text{ m}^3$	31%
Vacuum	$1.56 \times 10^{38} \text{ m}^3$	69%
Total	$2.26 \times 10^{38} \text{ m}^3$	

$$R_{\text{initial}} = (V_{\text{initial}} / 2\pi^2)^{1/3} \approx 15 \text{ AU}$$

At this stage, baryons and vacuum are packed together as QGVP — undifferentiated. The 31/69 split is retrospective: it describes what the components will become after phase separation.

6.3 Baryon census with top-heavy IMF

Approximately 3×10^{12} galaxies populate the full S^3 . With different IMF scenarios:

Scenario	BH mass/galaxy	Ω_b	R_{initial}
Conservative	$10^{11} M_{\odot}$	0.099	13.9 AU
Moderate	$5 \times 10^{11} M_{\odot}$	0.216	14.6 AU
Aggressive	$1.5 \times 10^{12} M_{\odot}$	0.511	16.0 AU

The initial radius is robustly 14–16 AU across all scenarios, because vacuum energy (57–69% of the initial volume) dominates the budget.

6.4 Self-consistency: the Friedmann integral

The time from the K-Limit to today, computed numerically:

t = 13.48 Gyr

Observed: 13.5–13.8 Gyr. Agreement: 2.3%.

Epoch	T (K)	t	R
K-Limit (QGVP)	1.8×10^{12}	0	15 AU
QGVP → matter + vacuum + CMB	1.5×10^{12}	24 μs	18 AU
Nucleosynthesis	10^{10}	3 min	2700 AU
Matter-radiation equality	9000	54 kyr	3×10^9 AU
Recombination	3000	370 kyr	9×10^9 AU
Today	2.725	13.48 Gyr	69 Gly

7. The Triangle

A → B. The K-Limit determines the initial state (QGVP on S^3). At the QCD phase transition, QGVP separates into matter + vacuum + radiation. The CMB is the latent heat of this separation. $\eta = \alpha^4$.

B → C. The changed photon production mechanism recalibrates Ω_b upward. More baryons — combined with top-heavy IMF producing trillions of compact remnants — account for the missing mass. The spatial profile is provided by vacuum capture.

C $\rightarrow$ A. No dark matter particles means no physics beyond the Standard Model at the initial state. The K-Limit is sufficient: QCD + GR + Starobinsky. Self-consistent only if no exotic particles are needed — which is true only if B and C hold.

Remove any vertex — the other two collapse.

8. Why No One Calculated This Before

The coherence triangle spans four departments that do not communicate.

Nuclear physics (σ -terms, QCD phase transition) provides α and the K-Limit. Stellar astrophysics (IMF, compact remnants) provides the baryon census. Observational cosmology (CMB peaks, BAO) provides the calibration of Ω_b — model-dependent but treated as fixed. General relativity (trace-free formulation, R^2 -term) provides the geometric framework.

No specialist sees all four. The nuclear physicist does not question Ω_b . The stellar astrophysicist does not recalibrate CMB peaks. The cosmologist does not count dead stars. The relativist does not compute σ -terms.

The coherence triangle requires all four simultaneously. It cannot be discovered within any single department.

9. Falsifiable Predictions

From Vertex A: (a) Tidal Love numbers $k_2 > 0$ above TOV limit (Einstein Telescope, LISA). (b) $\Omega_K < 0$ (CMB-S4, ± 0.001). (c) $r \approx 0.004$ (LiteBIRD).

From Vertex B: (d) $\eta = \alpha^4$ — correlated with lattice refinements of $\sigma_{\pi N}$. (e) Non-standard CMB spectral distortions from entropy injection (PIXIE). (f) Recalibrated $\Omega_b > 0.049$ from modified Boltzmann code.

From Vertex C: (g) No dark matter particles (LZ, XENONnT, PandaX). (h) Rotation curve truncation at vacuum capture radius (SKA). (i) Void galaxies retain standard $M_{\text{total}}/M_{\text{visible}}$ ratio (Euclid). (j) Excess stellar-mass black hole population (LISA).

The triangle as a whole: (k) All vertices simultaneously consistent — or none. Failure of any one collapses all three.

10. Conclusion

Three problems — the initial state, the CMB, and the missing mass — are one problem: the relationship between vacuum energy and gravity.

Zel'dovich answered: identity. The LGQV framework answers: local coupling through the chiral condensate.

The initial state is the K-Limit — QGVP on S^3 . The CMB is the latent heat of vacuum birth — the energy released when QGVP separates into matter and vacuum at the QCD phase transition. The missing mass is dead stars (from top-heavy IMF) plus vacuum profile (from σ -term shift in the galactic potential).

No vertex can be proven alone. All three support each other. The triangle stands or falls as a whole.

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